

requirements. Certain companies are offering induction lamps even at 13.56 MHz. This is within the industrial/scientific/medical (ISM) band, in which the FCC allows unlimited radio-frequency emissions. However, to meet FCC's residential requirements, the lamp must be shielded to keep emissions under control.

Colour temperature

Induction lamps are virtually maintenance free. These offer crisp white light with 80+ colour rendering index (CRI) and a choice of 3K, 4K, 5K and 6K colour temperatures. The high CRI light makes colours look brighter, more vibrant and more attractive. The lamps produce up to 80 lumens of light for each watt of energy. The 80LPW efficacy makes these as energy-efficient as CRI metal-halide systems. Induction lamps offer high reliability and instant on/off, with less thermal dissipation. The lamps' amalgam fill technology and the heat conduction rod in the centre create a stable light output over a wide range of ambient temperatures. These can start even at temperatures as low as -40°F.

Life and Efficacy Comparison of Different Lamps

Economic life*

Induction lamps 100,000 hours

Three-band fluorescent lamps 15 mm dia/high luminous flux

18,000 hours

Compact fluorescent lamps with EB

6,500 hours

Incandescent lamps (average service life)

1,000 hours

Luminous efficacy**

Induction lamps 90 lm/W

Three-band fluorescent lamps 15 mm dia/high luminous flux

88 lm/W

Compact fluorescent lamps

75 lm/W

Incandescent lamps

14 lm/W

Maximum luminous efficacy of lamps in lumens/watt (lm/W)

* Economic life is the length of time for which the lighting system delivers at least 70% of its initial luminous flux. Please note: in the case of fluorescent lamps, the basis used for measuring economic life is normally 50%. The same basis is used here.
** Maximum luminous efficacy of lamps in lumens/watt (lm/W)

Performance Comparison of Different Lamps

	LPW	Rate hour	CRI	Ignition time	Colour temp.	Major drawback
Incandescent	11-15	1.5-5K	40	Instant		Very inefficient-short life
Mercury vapour	13-48	12-24K		2-15 min	4000	Inefficiency
HPS	45-110	12-24K	25	2-15 min	2000	Low CRI
LPS	80-180	10-18K	0	2-15 min	1800	Low CRI
Metal halide	60-100	10-15K	75	2-15 min	3000-4300	High maintenance
Induction lamp	61-76	100-120K	82	0.05 sec	2600-6500	Relatively high initial cost

Disadvantages

1. Far higher initial cost (over ten times the cost of conventional HID lamps)
2. Currently limited in wattage. For example, the Philips QL and Osram Sylvania ICETRON range in initial lumen output of 3500 to 12,000, limiting their applicability to conventional metal-halide outdoor lighting designs of up to 150 watts
3. Physically larger than HID lamps, making them most appropriate for installation in large fixtures such as decorative post tops
4. Relatively limited selection

Due to the built-in pre-conditioner in the HF generator, which provides

a well stabilised internal supply voltage, the efficacy of the lamp system varies by less than 2 per cent as a result of mains voltage fluctuation. There is no noticeable effect (visual or measurable) on the colour performance (colour temperature, colour rendering, etc) due to supply voltage fluctuation.

Stray radiation and interference

Meticulous care design is required to generate a pure sine wave, as this waveform contains no harmonics. The bulb has a conductive fluorine-tin-

Benefits

1. High reliability and virtually maintenance-free operation
2. Lasting light output throughout lifetime
3. Economic, environment-friendly in situations with long burning hours
4. Restful non-fatiguing light
5. Automatic switch-off in case of lamp or power coupler failure
6. Suitable for security and safety lighting
7. More comfort
8. Useful light immediately after switch-on
9. Installation can be matched to lighting situations and requirements
10. Uniform colour maintained throughout life and the latest ones have the dimming facility that can increase energy savings and provide smooth light transition from ignition to full burn

oxide coating to screen undesirable radiation. Apart from visible light, the induction lamp system emits a certain amount of invisible infrared radiation. This radiation is extremely low, compared to HF fluorescent lamp systems.

Lamp life

Since there is no filament to burn out or electrodes to fail, an induction lamp can be turned on and off without affecting the lamp life. The product life is limited only by the lamp phosphor depreciation.

The life expectancy of an induction lamp system is almost 15 years of continuous operation of 4000 burning hours per year with an expected